Questions and results for my supervisor

ZuCo retrieval audit. Ziad Mohamed, 6 November 2026

Questions

1. Is a benchmark audit acceptable as my project outcome, even though decoding hasn't improved?
2. Is the paper still worth pursuing, given that similar public code exists? Should I contact those authors?
3. I've asked SemKey's authors for their outputs. If they say no, should I retrain the model? It needs a 52 GB download and a bigger GPU than mine.
4. Should I stop decoding work, or run one last test on the first-fixation result?
5. Which venue and deadline should I aim for?
6. When should I use my final test data? It can only be scored once.
7. Do we need to check anything in the ZuCo data terms before submitting?

Best results so far

Main point: EEG clearly beat noise only on a random split, where list position helps. On held-out list blocks, it did not beat noise. The top score, 28.18, is no better than noise. These are the best scores, and most are exploratory scores on reused data, so none is a final result.

Expected 10-way, chance 10%	Real EEG	Noise	Best with no EEG	Data
Same sentence across releases, nothing fitted	28.18	26.68	54.39 list position	48 sentences
CNN, random sentence split	23.10	9.88	53.94 neighbouring text	Validation, familiar readers
EEG predicts list position, then copies text from that position	15.41	11.93	55.10 neighbouring text	53 validation sentences
Held-out blocks, word-locked	14.55	16.25	26.05 gaze and word properties	255 reused sentences
New readers, word-locked	13.22	13.31		Same sentences, 15 scored readers
Held-out blocks, whole-sentence ridge	12.66	12.41	13.03 word counts and timing	Validation, 100 reused sentences
Held-out blocks, averaged across readers	11.25	12.13	12.07 timing	255 reused sentences

Word-locked scores use centred scoring, so they are not on exactly the same scale as the others. The 53.94 includes protected sentences and is on the CNN's own split; the 55.10 excludes them.

Other measures	Real EEG	Noise	Note
CNN, random split, 2-way (chance 50%)	66.42	50.25	Same run as the 23.10
Predicted vs true list position, correlation, ZuCo 1.0 / 2.0	0.80 / 0.88	0.54 / 0.69	Position, not meaning
Supplied word-band features, top-1 of 298 sentences (chance 0.34%)	12.30	10.74	Word count and duration alone: 17.77. Reused validation
Older test sets, best top-1, LaBraM on ZuCo 2.0, 344 candidates	0.41	0.51	Historical, not new confirmation
Word class, top-1, training-reader CV	3.01	3.00	Word task, not sentence accuracy. Missing-data pattern alone: 3.33

What the audit found

- **GLIM's public outputs (not my model):** 21.46% at 10-way across tasks, but 9.73% to 14.25% within one task. At 2-way, movie reviews vs biographies scored 87.36% to 93.57%, and biography tasks vs each other 51.89% to 60.66%.
- **Unchecked leads:** first-fixation waveform 12.81% vs noise 9.72%, and time-binned same-sentence matching 12.75% vs noise 5.70%. Neither has an interval, and both could be eye-movement traces.
- **Still unused:** the v2 test blocks have not been scored.

Scores at 2, 4, 10 and 24 way

Expected N-way accuracy, in percent. Compare real EEG with the noise row beneath it, not with other tests, as the sentences and setups differ. All of these use exploratory or reused data.

Test	2-way	4-way	10-way	24-way
Chance	50.00	25.00	10.00	4.17
CNN, random split (v1): real EEG	66.42	42.59	23.10	12.56
Matched noise	50.25	24.73	9.88	4.11
Ridge, random split (v1): real EEG	60.45	35.30	16.44	7.54
Matched noise	56.01	29.44	12.08	5.21
CNN, held-out blocks (v2): real EEG	50.00	25.09	10.02	4.25
Matched noise	49.80	24.82	9.72	3.94
Ridge F, held-out blocks (v2): real EEG	53.22	28.85	12.66	5.73
Matched noise	53.62	28.75	12.41	5.46
Word counts and timing, no EEG	53.44	29.77	13.03	4.76
Position test (v4): real EEG	55.48	31.47	15.41	7.77
Matched noise	50.89	26.46	11.93	6.53
Same sentence across releases (v7): real EEG	73.40	49.75	28.18	16.29
Matched noise	68.77	45.61	26.68	16.68
Averaged across readers (v5), ZuCo 1.0: real EEG	49.45	25.27	10.17	3.74
Matched noise	49.94	24.75	8.80	2.52
Averaged across readers (v5), ZuCo 2.0: real EEG	49.32	24.28	10.17	5.05
Matched noise	48.61	23.90	9.64	4.80
Word-locked (v6), ZuCo 1.0: real EEG	45.71	19.16	6.10	2.07
Matched noise	48.72	22.22	6.95	2.03
Gaze and word properties, no EEG	60.75	36.40	18.38	8.39
Word-locked (v6), ZuCo 2.0: real EEG	40.83	18.49	6.74	2.53
Matched noise	42.22	18.90	6.54	2.57
Gaze and word properties, no EEG	66.52	40.55	18.07	7.22

v5 and v6 rows score against all scored sentences in a release, so they differ from the within-block scores on page 1 (11.25 and 14.55). The v6 rows pool models fitted on different folds, which may be why real EEG and noise fall below chance; this was not tested. 5-way appears only in an earlier word-pretraining analysis on training-reader CV: real EEG 19.53, noise 18.81, chance 20.